FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2022

TEST DATE: 06-03-2022

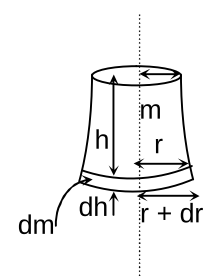
ANSWERS, HINTS & SOLUTIONS

Physics PART – A

SECTION – A

1. A

Sol. Stress at any point at distance h from the top is due to the weight of part above it



$$\frac{mg}{\pi r^2} = c \text{ (constant)}$$

$$m = \frac{c\pi r^2}{g}$$

$$dm = \frac{\pi c 2rdr}{g}$$

$$\rho\pi r^2 dh = \frac{2\pi c}{g} rdr$$

$$\int_0^H dh = \frac{2c}{\rho g}\int_{R_1}^{R_2}\frac{dr}{r}$$

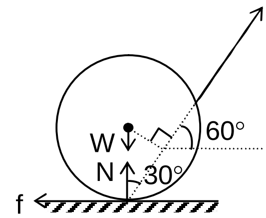
$$H = \frac{2c}{\rho g}\ell n\left(\frac{R_2}{R_1}\right)$$

$$\frac{R_2}{R_1} = e^{+\frac{\rho gH}{2c}}$$

$$\frac{R_1}{R_2} = e^{-\frac{\rho gH}{2c}}$$

2. D

Sol. The disc is in complete equilibrium



$\frac{F}{2} = f$ (Rotational equilibrium and translation equilibrium)

$F\frac{\sqrt{3}}{2} + N = W$ (Translational equilibrium along vertical direction)

Also $f \le \mu N$

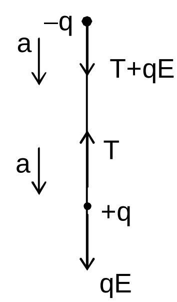
$$\frac{F}{2} \le \mu\left(-\frac{F\sqrt{2}}{2} + w\right)$$

$$\frac{F}{2}\left[1+\sqrt{3}\mu\right] \le w\mu$$

$$F \le \frac{2w\mu}{1+\sqrt{3}\mu}$$

3. D

Sol.



$$\frac{T+qE}{2m} = \frac{qE-T}{m}$$

T + qE = 2qE – 2T

$$T = \frac{qE}{3}$$

$$T = \frac{qkQR}{3(R^2+R^2)^{3/2}} = \frac{kQqR}{3(2R^2)^{3/2}}$$

$$T = \frac{kQqR}{3\left[2\sqrt{2}\right]R^3} = \frac{kQq}{6\sqrt{2}R^2}$$

4. C

Sol. $-\frac{dT}{dt} = k\left[T - T_S\right]$ also $k \propto T_S^3$

Case I : when $T_S = 20°C \Rightarrow k_1 = k$

Case II: when $T_S = 10°C \Rightarrow k_2 = k/8$

$$\frac{50-40}{5\text{ min}} = k\left[\frac{50+40}{2} - 20\right] \quad \ldots(i)$$

$$\frac{40-30}{t} = \frac{k}{8}\left[\frac{40+30}{2} - 10\right] \quad \ldots(ii)$$

From (i) and (ii) $\frac{t}{5\text{ min}} = 8$

$\Rightarrow$ t = 40 minutes

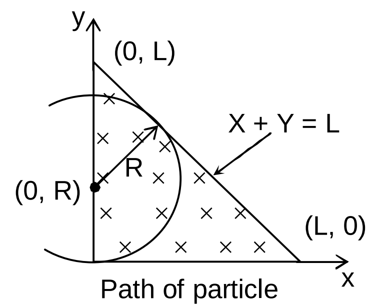
5. B

Sol. For v to be maximum the line x + y = L must be tangent to the circle

$x^2 + (y - R)^2 = R^2$

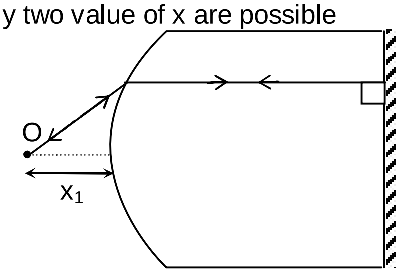
Hence $R = \left(\sqrt{2} - 1\right)L$

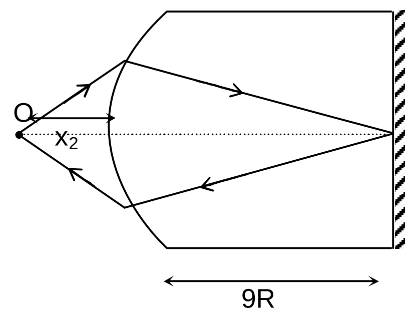
$V_{max} = (\sqrt{2} - 1)\dfrac{LBq}{m}$



6. A

Sol. Only two value of x are possible





$$\frac{3}{2(\infty)} - \frac{1}{-x_1} = \frac{\left(\frac{3}{2} - 1\right)}{R}$$

$x_1 = 2R$

$x_1 + x_2 = 5R$

$$\frac{3}{2(9R)} - \frac{1}{-x_2} = \frac{\left(\frac{3}{2} - 1\right)}{R}$$

$x_2 = 3R$

7. D

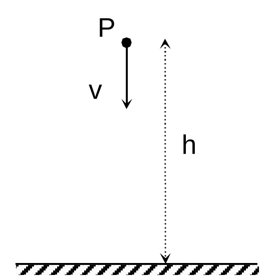
Sol. To reach at point P after collision, velocity of ball just after collision must be greater than or equal to $\sqrt{2gh}$

$\therefore$ speed of ball before collision is greater than or equal to $\dfrac{\sqrt{2gh}}{e}$

According to of conservation of mechanical energy

$$\frac{1}{2}mv_{min}^2 + mgh = \frac{1}{2}m\left(\frac{\sqrt{2gh}}{e}\right)^2$$

$$v_{min} = \sqrt{2gh\left(\frac{1}{e^2} - 1\right)}$$



8. B

Sol. $$\frac{1}{2}\left(\frac{2}{1000 + 100}\right) = \left(\frac{S}{100 + S}\right)\frac{2}{\left(1000 + \frac{100S}{100 + S}\right)}$$

$S \approx 91\,\Omega$

9. C

Sol. $W = \int \vec{F} \cdot d\vec{r} = \int_{(0,0)}^{(1,1)} (y^2dx + 2xydy) = \int_{(0,0)}^{(1,1)} d(xy^2) = \left[xy^2\right]_{(0,0)}^{(1,1)} = 1$ Joule

The given force is conservative and work done by this force is some along any path.

10. A

Sol. For $\omega > \frac{1}{\sqrt{LC}} \Rightarrow x_L > x_C$

Hence I_2 leads I_1

11. A

Sol. $5k\Omega(i_1) = 60$ V, $i_1 = 12$ mA

$240V - i_3\ 10\ k\Omega = 60$ V, $i_3 = 18$ mA

$\therefore\ i_2 = 6mA$

12. D

Sol. There is no tendency of slipping as acceleration of both the blocks are same in the absence of friction

$\therefore$ f = 0 N

13. C

Sol. $v = \sqrt{\frac{\gamma RT}{M}}$

$v_0 = \sqrt{\frac{7RT}{5(2)}}$

$v_{mix} = \sqrt{\frac{\gamma_{mix}RT}{M_{mix}}}$

$M_{mix} = \frac{n[4] + n[32]}{2n} = 18$

$\frac{2n}{\gamma_{mix} - 1} = \frac{n}{\left(\frac{7}{5} - 1\right)} + \frac{n}{\left(\frac{5}{3} - 1\right)}$

$\Rightarrow\ \gamma_{mix} = \frac{3}{2}$

$v_{mix} = \sqrt{\frac{5}{42}}v_0$

14. C

Sol. PT^2 = constant

$\Rightarrow\ \frac{T^3}{V} = \text{constant}$

TdV = 3VdT

$\gamma = \frac{dV}{VdT} = \frac{3}{T}$

15. D

Sol. When terminal velocity is acquired, then

$f_B + f_v = W$

$$\sigma\frac{4}{3}\pi r^3 g + 6\pi r\eta r v_T = \rho\frac{4}{3}\pi r^3 g$$
$$v_T \propto r^2$$
$$f_v = 6\pi\eta r v_T$$
Rate at which heat is produced = $f_V(v_T) = 6\pi\eta r v_T^2 \propto r^5$

16. C

Sol. $-\frac{f_o}{f_e}\left[1+\frac{f_e}{D}\right] = \frac{110}{100}\left(-\frac{f_o}{f_e}\right)$

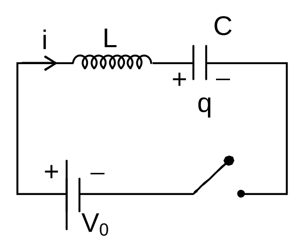
$\frac{f_e}{D} = \frac{1}{10}$

$f_e = \frac{25}{10}$ cm $\Rightarrow f_e = 2.5$ cm

L = $f_e + f_o$ (in normal adjustment)
10 cm = 2.5 cm + f_o
$\Rightarrow f_o = 7.5$ cm

17. B

Sol. $V_0 - \frac{Ldi}{dt} = \frac{q}{C}$(KVL) …(i)



Also, $i = \frac{dq}{dt}$ …(ii)

From (i) and (ii)

$\frac{d^2i}{dt^2} = -\frac{1}{LC}i \Rightarrow i = i_m\sin(\omega t)$

At t = 0, $L\frac{di}{dt} = V_0$

$Li_m\omega = V_0$

$i_m = \frac{V_0}{\omega L} = \frac{V_0}{L\frac{1}{\sqrt{LC}}} = V_0\sqrt{\frac{C}{L}}$

$i_m = 2\sqrt{\frac{8}{2}}\times 10^{-3}\text{ A} = 4\text{ mA}$

18. D

Sol. $\frac{\left(\frac{1}{3\mu F}\right)}{2\Omega} = \frac{\left(\frac{1}{C}\right)}{6\Omega} \Rightarrow C = 1\ \mu F$

19. B

Sol. $i = \frac{I\lambda eA\cos\theta}{hC} = 4\text{ A}$

20. A

Sol. $I_0 \propto \cos^2\left[(\mu-1)\left[\frac{\pi t}{\lambda}\right]\right]$

SECTION – B

21. 00008.00

Sol. $\frac{1}{2}mv_0^2 - \frac{GmM}{5R} = \frac{1}{2}mv^2 - \frac{GmM}{R}$ …(i) (conservation of mechanical energy)

$\frac{mv_0}{2}5R = mvR$ …(ii) (conservation of angular momentum about C)

from (i) and (ii)

$v_0 = \sqrt{\frac{32GM}{105R}} \Rightarrow k = 8$

22. 00002.00

Sol. $(2n+1)\frac{\lambda}{4} = 53.8\text{ cm} + e$ e = end correction

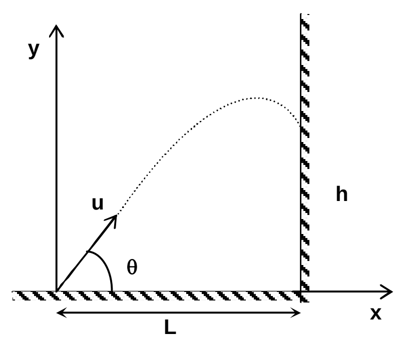
$(2n+3)\frac{\lambda}{4} = 75.8\text{ cm} + e$

$\therefore$ n = 2, λ = 44 cm $\Rightarrow$ e = 1.2 cm

e = 0.6 r, r = 2 cm

23. 00008.40

Sol. $y = x\tan\theta - \frac{gx^2}{2u^2}\sec^2\theta$



$h = L\tan\theta - \frac{gL^2}{2u^2}(1+\tan^2\theta)$

For h to be maximum

$\frac{dh}{d\theta} = 0 \Rightarrow u^2 = Lg\tan\theta$

$\therefore \tan\theta = \frac{(10\sqrt{2})^2}{8(10)} = \frac{5}{2}$

$h_{max} = 8\left(\frac{5}{2}\right) - \frac{10}{2}\frac{8^2}{200}\left(\frac{29}{4}\right) = 8.4\text{ m}$

24. 00006.00

Sol. $\lambda \propto \frac{1}{(z-1)^2}$

25. 00006.00

Sol. $\frac{\varepsilon^2(9)}{(9+r)^2} = \frac{\varepsilon^2(4)}{(4+r)^2}$

$\Rightarrow r = 6\Omega$

26. 00004.00

Sol. Energy in nth orbit = $-\frac{E_0}{n^2} = E_n$

$\therefore \frac{E_n - E_1}{e} = 15\text{ V} \Rightarrow \frac{15}{16}\frac{E_0}{e} = 15\text{ V}$

Also, $E_\infty - E_2 = \frac{E_0}{4} = 4\text{eV}$

27. 00032.00

Sol. $P_0V_0 = nRT_0$

$1.1P_0 \times 1.2V_0 = nRT$

$\Rightarrow T = 1.32\ T_0$

% change in T = $\frac{\Delta T}{T} \times 100 = 32\%$

28. 00000.00

Sol. After passes through 1st Polaroid, intensity of light becomes $\frac{I_0}{2}$ as it becomes polarised. Now the pass axis of 2nd poloroid is perpendicular to the first, hence the intensity after passes through 2nd polaroid is zero.

29. 00001.00

Sol. Upper half and lower half has the some focal length.

30. 00001.98

Sol. Applying energy conservation

$h\nu = 2m_0c^2 + E_1 + E_2$

$3\ \text{MeV} = 2[0.51]\text{MeV} + E_1 + E_2$

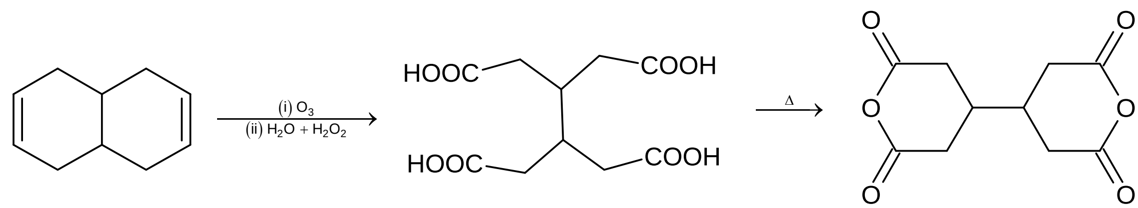
$E_1 + E_2 = 1.98$ MeV

Chemistry PART – B

SECTION – A

31. D

Sol.



32. C

Sol. II and III are more stable than I and IV due to complete octet. I is more stable than IV because in I, more electronegative atom has – ve charge.

33. D

Sol. (A) $\Delta S = +ve$ because in chain form more rotation is possible in C – C single bond.

(B) Ions have more solvated H_2O than H_2O hence when they form CH_3COOH and H_2O, more H_2O will be free. $\Delta S > 0$.

(C) Solid to gas entropy increases. $\Delta S > 0$.

(D) Two gas molecules are forming one molecule hence $\Delta S < 0$.

34. D

Sol.

	n	ℓ	$(n+\ell)$
6s	6	0	6
6p	6	1	7
4f	4	3	7
5d	5	2	7

Electrons are filled in increasing order of energy level.

$6s \rightarrow 4f \rightarrow 5d \rightarrow 6p$

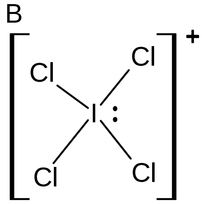
35. C

Sol. $O^-(g) + e \longrightarrow O^{-2}(g)$ $\Delta_{eg}H_2 = +ve$

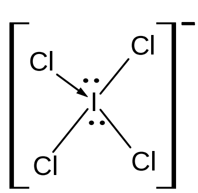
It is endothermic process.

36. B

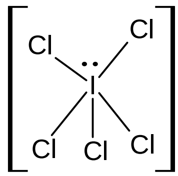
Sol.



See saw



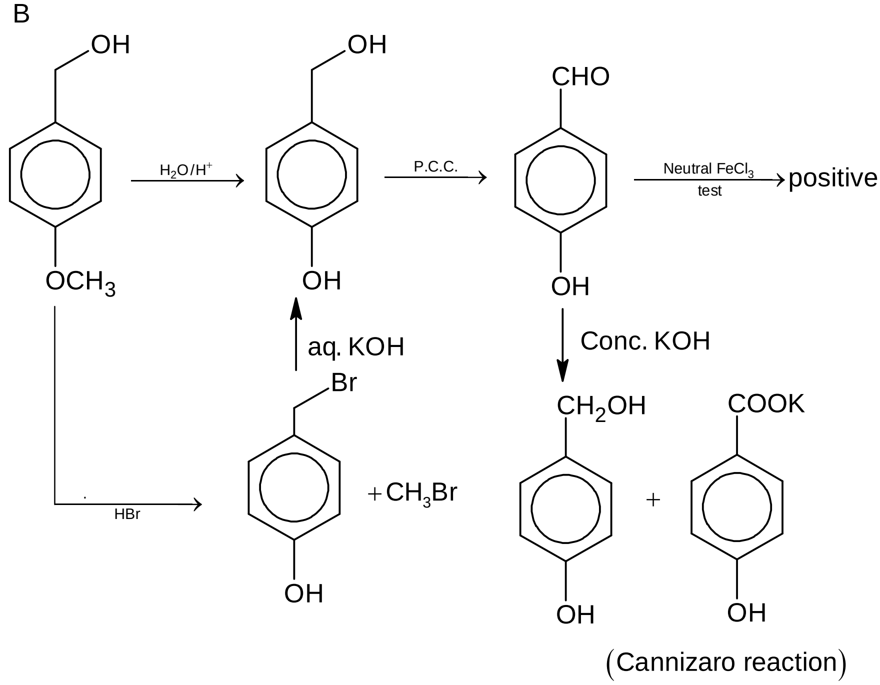
Square planer



Square pyramidal

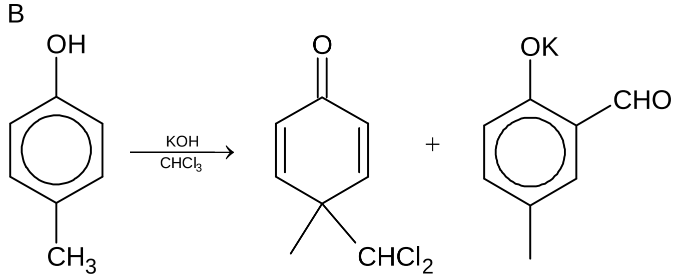
37. B

Sol.



38. B

Sol.



39. C

Sol. A, B, D are correct statements. Cellulose is a straight chain polysaccharide composed only of $\alpha-D-$Glucose. units.

40. A

Sol. $PV = nRT \qquad \left(1P_a.m^3 = 1\text{ Joule}\right)$

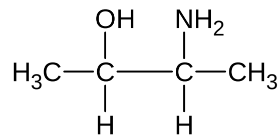
$50 \times 10 = (1.5 + y)R \times 2000$

$(1.5 + y)R = \frac{1}{4}$

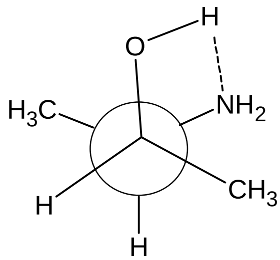
$1.5 + y = \frac{1}{4R} \Rightarrow \frac{3}{2} + y = \frac{1}{4R}$

$y = \frac{1}{4R} - \frac{3}{2} = \frac{1-6R}{4R}$

41. D

Sol. 3-amino-2-butanol 

It's Gauche form is most stable due to hydrogen bonding



42. D

Sol. $\alpha-D-\text{glucose} \rightleftharpoons \beta\text{-D-glucose}$

$$K_{eq} = \frac{[\beta]}{[\alpha]} = 1.4$$

$$1 + \frac{[\beta]}{[\alpha]} = 1.4 + 1$$

$$\frac{[\alpha]}{[\alpha]+[\beta]} = \frac{1}{2.4}$$

$$\%\text{ of } \alpha-\text{form} = \frac{1}{2.4} \times 100 = 41.67$$

43. A

Sol. B, C and D are wrong options.
Maximum limit of nitrae is 10 ppm in drinking water.
BOD value would have less than 5 ppm in clean water

44. C

Sol. $\left[Pt(NH_3)_4Br_2\right]Cl_2$ has 2 geometrical isomers
$\left[Pt(NH_3)_4ClBr\right]ClBr$ has 2 geometrical isomers
$\left[Pt(NH_3)_4Br_2\right]Cl_2$ has 2 geometrical isomers

45. D

Sol. Neoprene and Buna-N are synthetic rubber. Valcanised rubber is rubber which is formed by natural rubber and sulphur at proper temperature.

46. C

Sol. Melting of ice at $-10^{o}C$ is a non-spontaneous process. Hence, $\Delta G > 0$.

47. A

Sol. $NiCl_2(aq) \longrightarrow Ni^{+2}(aq) + 2Cl^{-}(aq)$
At Pt cathode reduction will take place
$Ni^{+2} + 2e \longrightarrow Ni(s)$

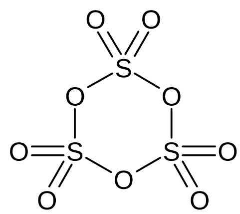
48. C

Sol. (C) is incorrect reaction

$$4Sn + 10HNO_3(dil.) \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

49. B

Sol. S_3O_9 is γ form of SO_3



50. D

Sol. Limonite $\rightarrow Fe_2O_3.3H_2O$

Zincite $\rightarrow ZnO$

Diaspore $\rightarrow Al_2O_3.H_2O$

Siderite $\rightarrow FeCO_3$

SECTION – B

51. 00001.50

Sol. $(-COOH)_n + nKOH \longrightarrow n(-COOK) + nH_2O$

Moles of acid × n = moles of KOH

$$\frac{10}{118} \times n = \frac{169.5 \times 1}{1000}$$

$n = 2$

$n = 2$ means two – COOH groups are present. Since, no loss of CO_2 take place, it means – COOH will be separated by carbon atoms.

Mass of left C and H in acid = 118 – 90 = 28

It will be by 2C and 4 hydrogen atoms.

Acid will be $HOOC - CH_2 - CH_2 - COOH$

$y = 6$

$x = 4$

$$\frac{4}{x} = \frac{6}{4} = \frac{3}{2} = 1.5$$

52. 00002.50

Sol. (iv) and (vi) are incorrect.

$$CH_2I_2 + Zn \longrightarrow \ddot{C}H_2 + ZnI_2$$

$$CH_3 - N = N - CH_3 \longrightarrow 2\dot{C}H_3 + N_2$$

$a = 5$

$b = 2$

$$\frac{a}{b} = \frac{5}{2} = 2.50$$

53. 00004.48
(Range 4.46 – 4.50)

Sol. $2NaClO_3 \xrightarrow{\Delta} 2NaCl + 3O_2$

$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

Moles of $O_2 = \frac{2}{3}$ moles of NaCl $= \frac{2}{3}$ moles of AgCl

Moles of AgCl $= \frac{2}{3}$ moles of O_2 $= \frac{2}{3} \times \left(\frac{1.5}{32}\right)$

Mass of AgCl $= \frac{2}{3} \times \frac{1.5}{32} \times 143.5\,\text{gm}$
= 4.484 gm
= 4.48 gm

54. 00002.25

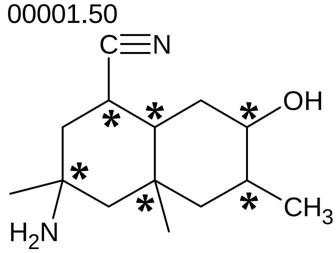
Sol. Correct statements are 4(i, iii, iv and v)

$x = 4$

$\frac{x+5}{4} = \frac{4+5}{4} = 2.25$

55. 00001.50

Sol.



Only four chiral carbon atoms have hydrogen atoms.
x = 6
y = 4

56. 00003.20

Sol. For He gas $P_{He} = \frac{2 \times 2}{5}$ $\left(\text{Use } P_1V_1 = P_2V_2\right)$

For H_2 gas $P_{H_2} = \frac{4 \times 3}{5}$ $\left(\text{Use } P_1V_1 = P_2V_2\right)$

$P_{total} = P_{He} + P_{H_2} = \frac{4}{5} + \frac{12}{5} = \frac{16}{5} = 3.20$ atm

Total pressure will be same in both chambers A and B.

57. 00008.55
(Range 8.50 – 8.60)

Sol.

	Cane sugar $+ H_2O(\ell) \longrightarrow$	Glucose +	Fructose
t=0	0.05 mole	0	0
t	0.05-0.05α	0.05α	0.05α

Total moles $= 0.05 - 0.05\alpha + 0.05\alpha + 0.05\alpha$

$= 0.05(1 + \alpha)$

$\Delta T_f = iK_f m$

$$0.279 = (1+\alpha)1.86 \times \left(\frac{0.05}{500/1000}\right)$$

$$(1+\alpha) = \frac{0.279 \times 5}{1.86 \times 0.5} = 1.5$$

$\alpha = 0.5$

Moles of cane sugar left $= 0.05 - 0.05\alpha$
$= 0.025$

Mass of cane sugar left $= 342 \times 0.025$
$= 8.55$ gm

58. 00001.50

Sol. Total number of surfactant molecules $= \left(\frac{100}{1000} \times \frac{50}{1000}\right) \times 6 \times 10^{23} = 3 \times 10^{21}$

Total area of surfactants $= 3 \times 10^{21} \times a^2 = 67.5\ \text{cm}^2$

(a = edge length)

$$a^2 = \frac{67.5}{3 \times 10^{21}} = 22.5 \times 10^{-21}$$

$a^2 = 2.25 \times 10^{-20}\ \text{cm}^2$

$a = 1.5 \times 10^{-10}\ \text{cm}$

$a = 1.50\ \text{Pm}$

59. 00001.25

Sol. $P(V - nb) = nRT$

For 1 mole $V = V_m = 5b$

$P(5b - b) = RT$

$$P = \frac{RT}{4b}$$

$$\frac{PV}{RT} = \frac{RT}{4b} \times \frac{V}{RT}$$

$$Z = \frac{V}{4b} \Rightarrow Z = \frac{5b}{4b}$$

Z = 1.25

60. 00002.62

Sol. $XeF_2 \longrightarrow Xe^+(g) + F^-(g) + F(g) \qquad \Delta H_r$

$\Delta H_r = 2\Delta H_{Xe-F} + \text{I.E}_{(Xe)} + \text{E.G.E.}_{(F)}$

$= 2 \times 34 + 279 - 85$

= 262

Mathematics

PART – C

SECTION – A

61. A

Sol. $\frac{11!}{4!\,4!\,2!\,3!} \times 2!\,3! = 69300$

62. A

Sol. $g'(x)$ changes sign at $\frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

63. B

Sol. $f(x)$ is even, has period 2, range is [0, 1] and symmetric about $x = 1$

64. D

Sol. $\int \tan^{-1}(xy)\,d(xy) = xy\tan^{-1}(xy) - \frac{1}{2}\ln\left(1+x^2y^2\right) + c$

65. C

Sol. Variance = $4 \times \left(\frac{100^2 - 1}{12}\right) = 3333$

66. C

Sol. Can be easily seen from truth table

67. C

Sol. $(a, a) \notin R$

if $(a, b) \in R \Rightarrow (b, a) \in R$

if $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \notin R$

68. A

Sol. Angle subtended by person at the centre in 5 seconds = $\frac{1}{2}$ radians

$\Rightarrow h = 2 \times 100\cos\left(\frac{\pi}{2} - \frac{1}{2}\cdot\frac{1}{2}\right)\tan(30^\circ) = \frac{200}{\sqrt{3}}\sin\left(\frac{1}{4}\right)$

69. B

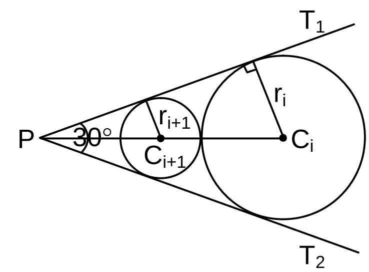
Sol. $\int \frac{\left(1 - 2xf(x) + 2x + f^2(x)\right)}{\left(1 - 2xf(x)\right)\left(x^2 + f^2(x) + 2xf(x)\right)}\,dx$

$= \int \frac{1 + \frac{2x + f^2(x)}{1 - 2xf(x)}}{\left(x + f(x)\right)^2}\,dx = \int \frac{1 + f'(x)}{\left(x + f(x)\right)^2}\,dx = \frac{-1}{x + f(x)} + c$

70. A

Sol. Either $m = 1, m' = -1$ or $m = -1, m' = 1$

71. D



Sol. $\dfrac{r_{i+1}}{r_i} = \dfrac{1-\sin\frac{\pi}{6}}{1+\sin\frac{\pi}{6}} = \dfrac{1}{3}$

$\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n} r_i = \dfrac{1}{1-\frac{1}{3}} = \dfrac{3}{2}$

72. D

Sol. Let Q(t) and $P\left(-t-\dfrac{2}{t}\right) \Rightarrow -t^2 - 2 = -1 \Rightarrow t^2 = -1$

So, point P does not exist

73. A

Sol. Coefficient of x^{674} in $(1+x)^{674}\,x^{674}\,(x+2)^{674} = 2^{674}$

74. B

Sol. Null matrix is the only matrix in set A and set B

75. A

Sol. $\tan x + 3\cot^3 x = \dfrac{4}{\sqrt{3}}\sin^2\left(\dfrac{\pi}{6}+x\right) \Rightarrow \tan x = \sqrt{3}$ and $\sin^2\left(\dfrac{\pi}{6}+x\right) = 1$

$\Rightarrow x = n\pi + \dfrac{\pi}{3}$ and $\dfrac{\pi}{6} + x = n\pi + \dfrac{\pi}{2}$; $x = n\pi + \dfrac{\pi}{3}$

76. C

Sol. $R = \dfrac{abc}{4\Delta} = \dfrac{4\cdot 5\cdot 6}{4\sqrt{\frac{15}{2}\left(\frac{7}{2}\right)\left(\frac{5}{2}\right)\left(\frac{3}{2}\right)}} = \dfrac{8}{\sqrt{7}}$

Area = πR^2

77. B

Sol. $2\tan^{-1}x = 2\cos^{-1}x$ is only satisfied for one value of x

78. D

Sol. dr's of normal vector of P_3 should be proportional to –1, –1, 1

So, a = b = –3

79. A

Sol. $\dfrac{\cos 2x}{\cos x \cos 3x} < 0$

Now solve using Wavy Curve Method to get $x \in \left(\dfrac{\pi}{6}\cdot\dfrac{\pi}{4}\right)$

80. C

Sol. As any vector in plane can be expressed as linear contribution of $\vec{a}$ and $\vec{b}$

$\Rightarrow \vec{a}, \vec{b}$ lie in plane and are non-collinear

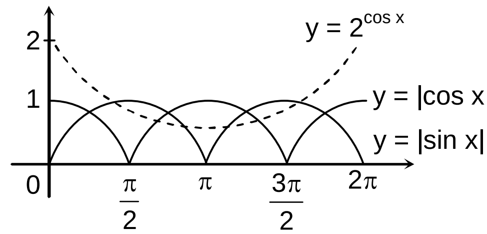
SECTION – B

81. 00269.00

Sol. $\frac{p}{q} = \frac{90}{180-1} = \frac{90}{179}$

82. 00006.00

Sol. Shown in the figure



83. 00008.00

Sol. A = {3, 4, 5, ..., } ; B = {....., –7, –6}

84. 04045.50

Sol. $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{2022 - 2023\,\text{cosec}^2 x}{\sin^{2022} x} dx = \left|\frac{\cot x}{\sin^{2022} x}\right|_{\frac{\pi}{6}}^{\frac{\pi}{3}} = -\frac{2^{2022}\left(3^{1012} - 1\right)}{3^{1011.5}}$

85. 00000.25

Sol. f(0) = 2 $\Rightarrow$ f(x + 2) + f(x – 2) = f(x) $\Rightarrow$ f(x) = f(x + 12)

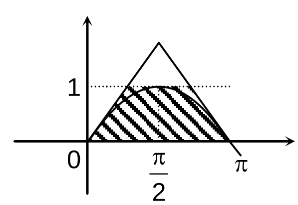
$\int_0^{12} f(x)dx = \frac{1}{4}\int_{-24}^{24} f(x - 36)dx$

86. 00248.00

Sol. Number of ways = ${}^{5000-4950+100-1}C_{100-1}$ = ${}^{149}C_{99}$

87. 00002.14

Sol. Bounded area = $\pi - 1$



88. 00003.46

Sol. $r = 2\sqrt{3}$

89. 00001.00

Sol. m = 1, n = 0

90. 00000.69

Sol. $T_r = \frac{1}{(2r-1)2r}$ (where r = 1, 2, ...)

$= \frac{1}{2r-1} - \frac{1}{2r}$ $\Rightarrow$ S = ln 2